# Lecture 20 — Transforming space

**Fri Oct 9**

Last updated: October 5, 2026.

**Boyd video (optional):** Lecture 19 — matrix–vector / transform examples · Lecture 23 — vectors as linear functions · ENGR108 playlist (Boyd)

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 7.1–7.2 and Lachniet Ch. 2. Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

## Learning objectives

By the end of class, you should be able to:

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

## Warm-up

Predict, without multiplying: where does $A = \mathrm{diag}(2, 1/2)$ send the point $(1, 1)$? Sketch the original and image roughly.

## Fill in

Many geometric maps on $\mathbb{R}^2$ (or $\mathbb{R}^3$) are matrix-vector products $y = Ax$.

**Finding the matrix.** The $j$th column of $A$ is what the map does to _____. So apply the geometry to $e_1, e_2$ (and $e_3$ in 3D) to read off columns.

Common $2 \times 2$ examples:

- **Scaling / dilation:** $A = \mathrm{diag}(s_1, s_2)$ stretches the axes by $s_1, s_2$.
- **Rotation** counterclockwise by $\theta$:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

- **Reflection** through a line through the origin; **shear** (e.g. $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$).

**Many points at once.** If columns of $X$ are points, then $AX$ applies the same map to every column.

**Change of coordinates (3D idea).** Body-fixed vs earth-fixed frames: $x_{\text{earth}} = p + Qx_{\text{body}}$, where columns of $Q$ are the body axes expressed in earth coordinates.

## Worked in class

**W1.** Build from $e_1, e_2$. Reflection through the $x_1$-axis sends $(1, 0) \mapsto (1, 0)$ and $(0, 1) \mapsto (0, -1)$. Write $A$. Check $A(2, 3)$.

**W2.** Rotation. For $\theta = 90^\circ = \pi/2$, write the rotation matrix and compute $A(1, 0)$ and $A(0, 1)$. Does that match "90° counterclockwise"?

**W3.** Many points. Let $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (three corners of a right triangle). With $A = \text{diag}(2, 1)$, compute $AX$ and describe the image.

**W4 (time permitting).** Shear $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Where do $(1, 0)$ and $(0, 1)$ go? Sketch the unit square's image.

## Your turn

**Y1.** Reflection through the $x_2$-axis: write $A$ by tracking $e_1$ and $e_2$, then compute $A(3, -1)$.

**Y2.** Rotation by 180°: write $A$ and apply it to $(2, 1)$.

**Y3.** True or false: every $2 \times 2$ matrix represents a rotation of the plane.

## Answers (Your turn)

Y1. $e_1 \mapsto (-1, 0)$, $e_2 \mapsto (0, 1)$, so $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. $A(3, -1) = (-3, -1)$.

Y2. $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$. $A(2, 1) = (-2, -1)$.

Y3. **False.** Scaling, shear, projections, and many other maps are also $2 \times 2$ matrices; rotations are a special family.